Linear Programming:

Reference: Operation Research by Hamdy A. Taha, 10th edition.

Problem-2.1-1 and 2.1-2 (Page-45-52), problem 2.7 , 2.8, 2.9, 2.13 (page 77, 78).

CHAPTER 2

Modeling with Linear Programming

Real-Life Application—Frontier Airlines Purchases Fuel Economically

The fueling of an aircraft can take place at any of the stopovers along a flight route. Fuel price varies among the stopovers, and potential savings can be realized by tankering (loading) extra fuel at a cheaper location for use on subsequent flight legs. The disadvantage is that the extra weight of tankered fuel will result in higher burn of gasoline. Linear programming (LP) and heuristics are used to determine the optimum amount of tankering that balances the cost of excess burn against the savings in fuel cost. The study, carried out in 1981, resulted in net savings of about $350,000 per year. With the significant rise in the cost of fuel, many airlines are using LP-based tankering software to purchase fuel. Details of the study are given in Case 1, Chapter 26 on the website.

2.1 TWO-VARIABLE LP MODEL

This section deals with the graphical solution of a two-variable LP. Though two-variable problems hardly exist in practice, the treatment provides concrete foundations for the development of the general simplex algorithm presented in Chapter 3.

Example 2.1-1 (The Reddy Mikks Company)

Reddy Mikks produces both interior and exterior paints from two raw materials, $M1$ and $M2$. The following table provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	Exterior paint	*Interior paint*	
Raw material, $M1$	6	4	24
Raw material, $M2$	1	2	6
Profit per ton ($1000)	5	4	

The daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton. Also, the maximum daily demand for interior paint is 2 tons.

Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.

All OR models, LP included, consist of three basic components:

1. **Decision variables** that we seek to determine.
2. **Objective** (goal) that we need to optimize (maximize or minimize).
3. **Constraints** that the solution must satisfy.

The proper definition of the decision variables is an essential first step in the development of the model. Once done, the task of constructing the objective function and the constraints becomes more straightforward.

For the Reddy Mikks problem, we need to determine the daily amounts of exterior and interior paints to be produced. Thus the variables of the model are defined as:

$$x_1 = \text{Tons produced daily of exterior paint}$$

$$x_2 = \text{Tons produced daily of interior paint}$$

The goal of Reddy Mikks is to *maximize* (i.e., increase as much as possible) the total daily profit of both paints. The two components of the total daily profit are expressed in terms of the variables x_1 and x_2 as:

$$\text{Profit from exterior paint} = 5x_1 \text{ (thousand) dollars}$$

$$\text{Profit from interior paint} = 4x_2 \text{ (thousand) dollars}$$

Letting z represent the total daily profit (in thousands of dollars), the objective (or goal) of Reddy Mikks is expressed as

$$\text{Maximize } z = 5x_1 + 4x_2$$

Next, we construct the constraints that restrict raw material usage and product demand. The raw material restrictions are expressed verbally as

$$\begin{pmatrix}\text{Usage of a raw material}\\ \text{by both paints}\end{pmatrix} \leq \begin{pmatrix}\text{Maximum raw material}\\ \text{availability}\end{pmatrix}$$

The daily usage of raw material $M1$ is 6 tons per ton of exterior paint and 4 tons per ton of interior paint. Thus,

$$\text{Usage of raw material } M1 \text{ by both paints} = 6x_1 + 4x_2 \text{ tons/day}$$

In a similar manner,

$$\text{Usage of raw material } M2 \text{ by both paints} = 1x_1 + 2x_2 \text{ tons/day}$$

The maximum daily availabilities of raw materials $M1$ and $M2$ are 24 and 6 tons, respectively. Thus, the raw material constraints are:

$$\begin{aligned} 6x_1 + 4x_2 &\leq 24 \quad \text{(Raw material } M1\text{)} \\ x_1 + 2x_2 &\leq 6 \quad \text{(Raw material } M2\text{)} \end{aligned}$$

The first restriction on product demand stipulates that the daily production of interior paint cannot exceed that of exterior paint by more than 1 ton, which translates to:

$$x_2 - x_1 \leq 1 \quad \text{(Market limit)}$$

The second restriction limits the daily demand of interior paint to 2 tons—that is,

$$x_2 \le 2 \quad \text{(Demand limit)}$$

An implicit (or "understood-to-be") restriction requires (all) the variables, x_1 and x_2, to assume zero or positive values only. The restrictions, expressed as $x_1 \ge 0$ and $x_2 \ge 0$, are referred to as **nonnegativity constraints.**

The complete Reddy Mikks model is

$$\text{Maximize } z = 5x_1 + 4x_2$$

subject to

$$\begin{aligned} 6x_1 + 4x_2 &\le 24 \quad (1) \\ x_1 + 2x_2 &\le \ \ 6 \quad (2) \\ -x_1 + x_2 &\le \ \ 1 \quad (3) \\ x_2 &\le \ \ 2 \quad (4) \\ x_1, x_2 &\ge \ \ 0 \quad (5) \end{aligned}$$

Any values of x_1 and x_2 that satisfy *all* five constraints constitute a **feasible solution.** Otherwise, the solution is **infeasible.** For example, the solution $x_1 = 3$ tons per day and $x_2 = 1$ ton per day is feasible because it does not violate *any* of the five constraints;a result that is confirmed by using substituting $(x_1 = 3, x_2 = 1)$ in the left-hand side of each constraint. In constraint (1), we have $6x_1 + 4x_2 = (6 \times 3) + (4 \times 1) = 22$, which is less than the right-hand side of the constraint $(= 24)$. Constraints 2 to 5 are checked in a similar manner (verify!). On the other hand, the solution $x_1 = 4$ and $x_2 = 1$ is infeasible because it does not satisfy at least one constraint. For example, in constraint (1), $(6 \times 4) + (4 \times 1) = 28$, which is larger than the right-hand side $(= 24)$.

The goal of the problem is to find the **optimum,** the best *feasible* solution that maximizes the total profit z. First, we need to show that the Reddy Mikks problem has an *infinite* number of feasible solutions, a property that is shared by all nontrivial LPs. Hence the problem cannot be solved by enumeration. The graphical method in Section 2.2 and its algebraic generalization in Chapter 3 show how the optimum can be determined in a finite number of steps.

Remarks. The objective and the constraint function in all LPs must be linear. Additionally, all the parameters (coefficients of the objective and constraint functions) of the model are known with certainty.

2.2 GRAPHICAL LP SOLUTION

The graphical solution includes two steps:

1. Determination of the feasible solution space.
2. Determination of the optimum solution from among all the points in the solution space.

The presentation uses two examples to show how maximization and minimization objective functions are handled.

2.2.1 Solution of a Maximization Model

Example 2.2-1

This example solves the Reddy Mikks model of Example 2.1-1.

Step 1. *Determination of the Feasible Solution Space:*

First, consider the nonnegativity constraints $x_1 \geq 0$ and $x_2 \geq 0$. In Figure 2.1, the horizontal axis x_1 and the vertical axis x_2 represent the exterior- and interior-paint variables, respectively. Thus, the nonnegativity constraints restrict the variables to the first quadrant (above the x_1-axis and to the right of the x_2-axis).

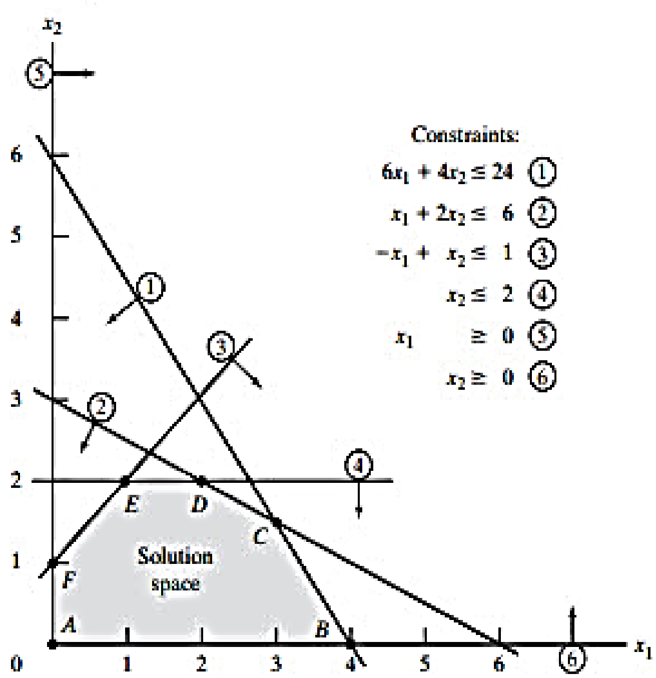
To account for the remaining four constraints, first replace each inequality with an equation, and then graph the resulting straight line by locating two distinct points. For example, after replacing $6x_1 + 4x_2 \leq 24$ with the straight line $6x_1 + 4x_2 = 24$, two distinct points are determined by setting $x_1 = 0$ to obtain $x_2 = \frac{24}{4} = 6$ and then by setting $x_2 = 0$ to obtain $x_1 = \frac{24}{6} = 4$. Thus the line $6x_1 + 4x_2 = 24$ passes through $(0, 6)$ and $(4, 0)$, as shown by line (1) in Figure 2.1.

Next, consider the direction ($>$ or $<$) of the inequality. It divides the (x_1, x_2) plane into two half-spaces, one on each side of the graphed line. Only one of these two halves satisfies the inequality. To determine the correct side, designate any point *not* lying on the straight line as a *reference point*. If the chosen reference point satisfies the inequality, then its side is feasible; otherwise, the opposite side becomes the feasible half-space.

The origin $(0, 0)$ is a convenient reference point and should always be used so long as it does not lie on the line representing the constraint. This happens to be true for all the constraints of this example. Starting with the constraint $6x_1 + 4x_2 \leq 24$,

FIGURE 2.1

Feasible space of the Reddy Mikks model



substitution of $(x_1, x_2) = (0, 0)$ automatically yields zero for the left-hand side. Since it is less than 24, the half-space containing $(0, 0)$ is feasible for inequality (1), as the direction of the arrow in Figure 2.1 shows. A similar application of the reference-point procedure to the remaining constraints produces the **feasible solution space** $ABCDEF$ in which all the constraints are satisfied (verify!). All points outside the boundary of the area $ABCDEF$ are infeasible.

Step 2. *Determination of the Optimum Solution:*
The number of solution points in the feasible space $ABCDEF$ in Figure 2.1 is *infinite*, clearly precluding the use of exhaustive enumeration. A systematic procedure is thus needed to determine the optimum solution.

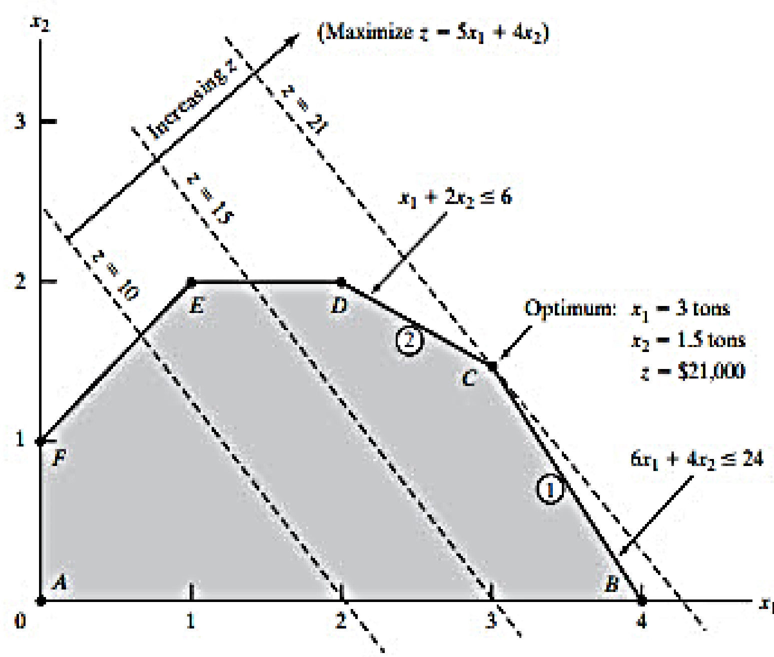
First, the direction in which the profit function $z = 5x_1 + 4x_2$ increases (recall that we are *maximizing* z) is determined by assigning arbitrary *increasing* values to z. In Figure 2.2, the two lines $5x_1 + 4x_2 = 10$ and $5x_1 + 4x_2 = 15$ corresponding to (arbitrary) $z = 10$ and $z = 15$ depict the direction in which z increases. Moving in that direction, the optimum solution occurs at C because it is the feasible point in the solution space beyond which any further increase will render an infeasible solution.

The values of x_1 and x_2 associated with the optimum point C are determined by solving the equations associated with lines (1) and (2):

$$6x_1 + 4x_2 = 24$$
$$x_1 + 2x_2 = 6$$

FIGURE 2.2

Optimum solution of the Reddy Mikks model



The solution is $x_1 = 3$ and $x_2 = 1.5$ with $z = (5 \times 3) + (4 \times 1.5) = 21$. This calls for a daily product mix of 3 tons of exterior paint and 1.5 tons of interior paint. The associated daily profit is \$21,000.

Remarks. In practice, a typical LP may include hundreds or even thousands of variables and constraints. Of what good then is the study of a two-variable LP? The answer is that the graphical solution provides a key result: *The optimum solution of an LP, when it exists, is always associated with a* ***corner point*** *of the solution space, thus limiting the search for the optimum from an infinite number of feasible points to a finite number of corner points.* This powerful result is the basis for the development of the general algebraic *simplex method* presented in Chapter 3.[1]

2.2.2 Solution of a Minimization Model

Example 2.2-2 (Diet Problem)

Ozark Farms uses at least 800 lb of special feed daily. The special feed is a mixture of corn and soybean meal with the following compositions:

	lb per lb of feedstuff		
Feedstuff	*Protein*	*Fiber*	Cost (\$/lb)
Corn	.09	.02	.30
Soybean meal	.60	.06	.90

The dietary requirements of the special feed are at least 30% protein and at most 5% fiber. The goal is to determine the daily minimum-cost feed mix.

The decision variables of the model are:

$$x_1 = \text{lb of corn in the daily mix}$$

$$x_2 = \text{lb of soybean meal in the daily mix}$$

The objective is to minimize the total daily cost (in dollars) of the feed mix—that is,

$$\text{Minimize } z = .3x_1 + .9x_2$$

[1]To reinforce this key result, use TORA to verify that the optimum of the following objective functions of the Reddy Mikks model (Example 2.1-1) will yield the associated corner points as defined in Figure 2.2 (click View/Modify Input Data to modify the objective coefficients and re-solve the problem graphically):

(a) $z = 5x_1 + x_2$ (optimum: point B in Figure 2.2)

(b) $z = 5x_1 + 4x_2$ (optimum: point C)

(c) $z = x_1 + 3x_2$ (optimum: point D)

(d) $z = x_2$ (optimum: point D or E, or any point inbetween—see Section 3.5.2)

(e) $z = -2x_1 + x_2$ (optimum: point F)

(f) $z = -x_1 - x_2$ (optimum: point A)

The constraints represent the daily amount of the mix and the dietary requirements. Ozark Farms needs at least 800 lb of feed a day—that is,

$$x_1 + x_2 \geq 800$$

The amount of protein included in x_1 lb of corn and x_2 lb of soybean meal is $(.09x_1 + .6x_2)$ lb. This quantity should equal at least 30% of the total feed mix $(x_1 + x_2)$ lb—that is,

$$.09x_1 + .6x_2 \geq .3(x_1 + x_2)$$

In a similar manner, the fiber requirement of at most 5% is represented as

$$.02x_1 + .06x_2 \leq .05(x_1 + x_2)$$

The constraints are simplified by moving the terms in x_1 and x_2 to the left-hand side of each inequality, leaving only a constant on the right-hand side. The complete model is

$$\text{Minimize } z = .3x_1 + .9x_2$$

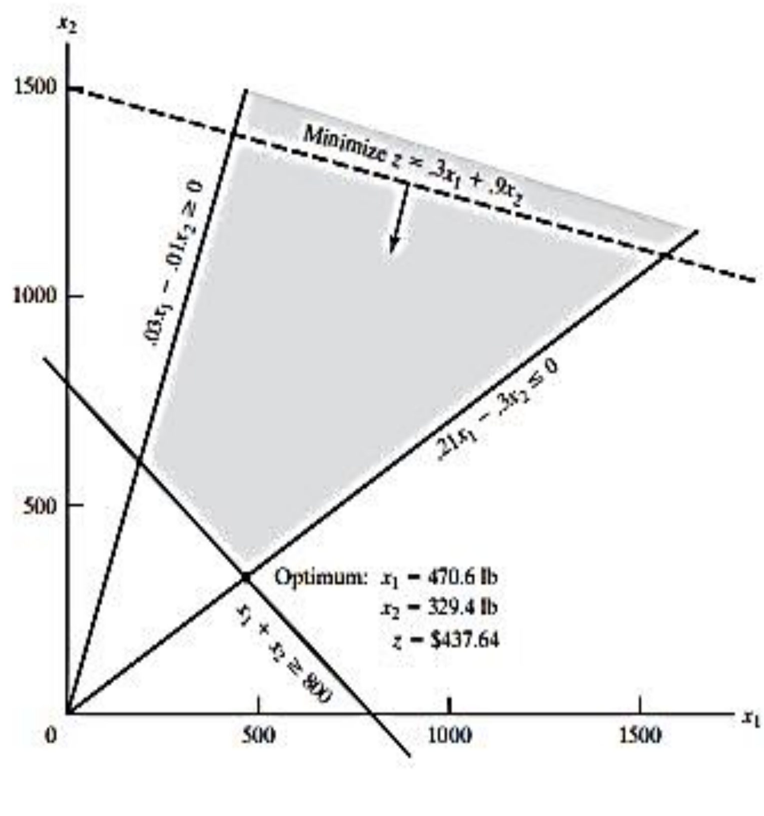
subject to

$$\begin{aligned} x_1 + \quad x_2 &\geq 800 \\ .21x_1 - .30x_2 &\leq 0 \\ .03x_1 - .01x_2 &\geq 0 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Figure 2.3 provides the graphical solution of the model. The second and third constraints pass through the origin. Thus, unlike the Reddy Mikks model of Example 2.2-1, the determination of

FIGURE 2.3

Graphical solution of the diet model



the feasible half-spaces of these two constraints requires using a reference point other than (0, 0) [e.g., (100, 0) or (0, 100)].

Solution:

The model minimizes the value of the objective function by reducing z in the direction shown in Figure 2.3. The optimum solution is the intersection of the two lines $x_1 + x_2 = 800$ and $.21x_1 - .3x_2 = 0$, which yields $x_1 = 470.6$ lb and $x_2 = 329.4$ lb. The minimum cost of the feed mix is $z = .3 \times 470.6 + .9 \times 329.4 = \$437.64$ per day.

Remarks. One may wonder why the constraint $x_1 + x_2 \geq 800$ cannot be replaced with $x_1 + x_2 = 800$ because it would not be optimum to produce more than the minimum quantity. Although the solution of the present model did satisfy the equation, a more complex model may impose additional restrictions that would require mixing more than the minimum amount. More importantly, the weak inequality ($\geq$), by definition, implies the equality case, so that the equation ($=$) is permitted if optimality requires it. The conclusion is that one should not "preguess" the solution by imposing the additional equality restriction.

2.3 COMPUTER SOLUTION WITH SOLVER AND AMPL

In practice, where typical LP models may involve thousands of variables and constraints, the computer is the only viable venue for solving LP problems. This section presents two commonly used software systems: Excel Solver and AMPL. Solver is particularly appealing to spreadsheet users. AMPL is an algebraic modeling language that, like all higher-order programming languages, requires more expertise. Nevertheless, AMPL, and similar languages,[2] offers great modeling flexibility. Although the presentation in this section concentrates on LPs, both AMPL and Solver can handle integer and nonlinear problems, as will be shown in later chapters.

2.3.1 LP Solution with Excel Solver

In Excel Solver, the spreadsheet is the input and output medium for the LP. Figure 2.4 shows the layout of the data for the Reddy Mikks model (file *solverRM1.xls*). The top of the figure includes four types of information: (1) input data cells (B5:C9 and F6:F9), (2) cells representing the variables and the objective function (B13:D13), (3) algebraic definitions of the objective function and the left-hand side of the constraints (cells D5:D9), and (4) cells that provide (optional) explanatory names or symbols. Solver requires the first three types only. The fourth type enhances readability but serves no other purpose. The relative positioning of the four types of information on the

[2]Other known commercial packages include AIMMS, GAMS, LINGO, MPL, OPL Studio, and Xpress-Mosel.

2-1. For the Reddy Mikks model, construct each of the following constraints, and express it with a linear left-hand side and a constant right-hand side:

*(a) The daily demand for interior paint exceeds that of exterior paint by *at least* 1 ton.

(b) The daily usage of raw material $M1$ in tons is *at most* 8 and *at least* 5.

*(c) The demand for exterior paint cannot be less than the demand for interior paint.

(d) The maximum quantity that should be produced of both the interior and the exterior paint is 15 tons.

*(e) The proportion of exterior paint to the total production of both interior and exterior paints must not exceed .5.

2-2. Determine the best *feasible* solution among the following (feasible and infeasible) solutions of the Reddy Mikks model:

(a) $x_1 = 1, x_2 = 2$.

(b) $x_1 = 3, x_2 = 1$.

(c) $x_1 = 3, x_2 = 1.5$.

(d) $x_1 = 2, x_2 = 1$.

(e) $x_1 = 2, x_2 = -1$.

***2-3.** For the feasible solution $x_1 = 1, x_2 = 2$ of the Reddy Mikks model, determine the unused amounts of raw materials $M1$ and $M2$.

2-4. Suppose that Reddy Mikks sells its exterior paint to a single wholesaler at a quantity discount. The profit per ton is \$5000 if the contractor buys no more than 5 tons daily and \$4300 otherwise. Express the objective function mathematically. Is the resulting function linear?

2-5. Determine the feasible space for each of the following independent constraints, given that $x_1, x_2 \geq 0$.

*(a) $-3x_1 + x_2 \leq 6$.

(b) $x_1 - 2x_2 \geq 5$.

(c) $2x_1 - 3x_2 \leq 12$.

(d) $x_1 - x_2 \leq 0$.

*(e) $-x_1 + x_2 \geq 0$.

2-6. Identify the direction of increase in z in each of the following cases:

*(a) Maximize $z = x_1 - x_2$.

(b) Maximize $z = -8x_1 - 3x_2$.

(c) Maximize $z = -x_1 + 3x_2$.

*(d) Maximize $z = -3x_1 + x_2$.

2-7. Determine the solution space and the optimum solution of the Reddy Mikks model for each of the following independent changes:

(a) The maximum daily demand for interior paint is 1.9 tons and that for exterior paint is at most 2.5 tons.

(b) The daily demand for interior paint is at least 2.5 tons.

(c) The daily demand for interior paint is exactly 1 ton higher than that for exterior paint.

(d) The daily availability of raw material $M1$ is at least 24 tons.

(e) The daily availability of raw material $M1$ is at least 24 tons, and the daily demand for interior paint exceeds that for exterior paint by at least 1 ton.

2-8. A company that operates 10 hrs a day manufactures two products on three sequential processes. The following table summarizes the data of the problem:

	Minutes per unit			
Product	*Process 1*	*Process 2*	*Process 3*	Unit profit
1	10	6	8	\$20
2	5	20	10	\$30

Determine the optimal mix of the two products.

***2-9.** A company produces two products, A and B. The sales volume for A is at least 80% of the total sales of both A and B. However, the company cannot sell more than 110 units of A per day. Both products use one raw material, of which the maximum daily availability is 300 lb. The usage rates of the raw material are 2 lb per unit of A, and 4 lb per unit of B. The profit units for A and B are \$40 and \$90, respectively. Determine the optimal product mix for the company.

2-10. Alumco manufactures aluminum sheets and aluminum bars. The maximum production capacity is estimated at either 800 sheets or 600 bars per day. The maximum daily demand is 550 sheets and 560 bars. The profit per ton is \$40 per sheet and \$35 per bar. Determine the optimal daily production mix.

***2-11.** An individual wishes to invest \$5000 over the next year in two types of investment: Investment A yields 5%, and investment B yields 8%. Market research recommends an allocation of at least 25% in A and at most 50% in B. Moreover, investment in A should be at least half the investment in B. How should the fund be allocated to the two investments?

2-12. The Continuing Education Division at the Ozark Community College offers a total of 30 courses each semester. The courses offered are usually of two types: practical and humanistic. To satisfy the demands of the community, at least 10 courses of each type must be offered each semester. The division estimates that the revenues of offering practical and humanistic courses are approximately \$1500 and \$1000 per course, respectively.

(a) Devise an optimal course offering for the college.

(b) Show that the worth per additional course is \$1500, which is the same as the revenue per practical course. What does this result mean in terms of offering additional courses?

2-13. ChemLabs uses raw materials I and II to produce two domestic cleaning solutions, A and B. The daily availabilities of raw materials I and II are 150 and 145 units, respectively. One unit of solution A consumes .5 unit of raw material I and .6 unit of raw material II. One unit of solution B uses .5 unit of raw material I and .4 unit of raw material II. The profits per unit of solutions A and B are \$8 and \$10, respectively. The daily demand for solution A lies between 30 and 150 units, and that for solution B between 40 and 200 units. Find the optimal production amounts of A and B.

2-14. In the Ma-and-Pa grocery store, shelf space is limited and must be used effectively to increase profit. Two cereal items, Grano and Wheatie, compete for a total shelf space of 60 ft^2. A box of Grano occupies .2 ft^2 and a box of Wheatie needs .4 ft^2. The maximum daily demands of Grano and Wheatie are 200 and 120 boxes, respectively. A box of Grano nets \$1.00 in profit and a box of Wheatie \$1.35. Ma-and-Pa thinks that because the unit profit of Wheatie is 35% higher than that of Grano, Wheatie should be allocated 35% more space than Grano, which amounts to allocating about 57% to Wheatie and 43% to Grano. What do you think?